

APPENDIX A

Optimal Hyper-parameter Settings for Models

Table 1. Logistic Regression

Logistic Regression	BOW	TFIDF	W2V
TR-Stem	min-df = 2	max_features= None	
TR-Lemma	min-df = 2	max_features= None	
ENG-Stem	min-df = 2	max_features= None	Max_length = 250
ENG-Lemma	min-df = 2	max_features= None	Max_length = 250

Table 2. Support Vector Classifier

SVC	BOW	TFIDF	W2V
TR-Stem	min-df = 3	max_features= 3500	
TR-Lemma	min-df = 3	max_features= 3500	
ENG-Stem	min-df = 3	max_features= 3500	Max_length = 250
ENG-Lemma	min-df = 3	max_features= 3500	Max_length = 250

Table 3. Multi-layer Perceptron

MLP	BOW	TFIDF	W2V
TR-Stem	min-df = 2	max_features= None	
TR-Lemma	# layers = 1 # nodes = 512	#layers = 1 # nodes = 512	
ENG-Stem	Epochs = 200 Learning rate = 0.0001	max_features= 2000 #layers = 1 #nodes = 128	Max_length = 250 #layers = 1 #nodes = 512
ENG-Lemma		max_features= 3500 #layers = 1 #nodes = 128	Max_length = 500 #layers = 1 #nodes = 512

* epochs and learning rate are the same for all cases

Table 4. Convolutional Neural Network

CNN	BOW	TFIDF	W2V
TR-Stem	min-df = 2 #layers = 1 #filters = 32 Kernel size = 3	max_features= None #layers = 1 #filters = 8 Kernel size = 3	
TR-Lemma	min-df = 2 #layers = 1 #filters = 8 Kernel size = 3		

ENG-Stem	min-df = 2 #layers = 1 #filters = 32 Kernel size = 3		Max_length = 500 #layers = 1 #filters = 32 Kernel size = 7
ENG-Lemma			Max_length = 500 #layers = 1 #filters = 8 Kernel size = 7

* epochs and learning rate are the same for all cases: 1000 and 0.0001 respectively

Table 5. Transformer

Transformer	TR-Stem	TR-Lemma	ENG-Stem	ENG-Lemma
Vocabulary Size	4000	3000	3000	2000
Max length	200	100	100	300
Embed Dimension	16	16	32	16
# heads	2	2	2	4
Ff dimensions	32	16	32	32

* epochs and learning rate are the same for all cases: 250 and 0.001 respectively

TOP-5 Model Runtimes

These runtimes are measured on a MacBook Laptop with M1-Pro chip with 8 cores and 16MB memory for top-5 models with their best hyper-parameter values and using 5-fold cross validation

Model	Runtime (Seconds)
LR-TFIDF-TR-LEMMA	45.15
LR-W2V-ENG-STEM	812.52
MLP-BOW-TR-LEMMA	103.12
MLP-BOW-TR-STEM	100.66
MLP-TFIDF-TR-STEM	125.64